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GENERAL CAPABILITY OF PARAMETRIC 3D BLADE DESIGN TOOL FOR TURBOMACHINERY

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ABSTRACT

A design tool for generating 3D blades for various turbomachinery applications using a parametric approach has been developed. The tool can create a variety of 3D blade geometries based on only a few basic parameters and limited interaction with a CAD system. A general approach for creating the blade geometries is implemented which makes it robust and easy to create different 3D blade shapes for various turbomachinery components. The geometric and aerodynamic parameters are used to create 2D airfoils and these airfoils are stacked on the desired stacking axis. The tool generates a specified number of 2D blade sections in a 3D Cartesian coordinate system. These sections can be lofted in a CAD package to obtain a solid 3D blade model, which has been demonstrated using Unigraphics-NX and Solidworks.

The geometry modeler can also be used for generating 3D blades with special features like bent tip, split tip and other concepts, which can be explored with minimum changes to the blade geometry. The use of control points for the definition of splines makes it easy to modify the blade shapes quickly and smoothly to obtain the desired blade model. Blade shapes for axial turbomachines, radial turbomachines and wind turbines are generated to show the general capability of the tool. Other novel blade shapes are also shown which shows the full utility of this tool when integrated with CAD. The executable of the code that generates sections is freely available on the web.

NOMENCLATURE

a	Offset value
b	Intermediate extruded blade scale factor
i	Incidence angle
L	Reference length
m	Meridional coordinate
mm	Millimeters
m'	Normalized Meridional coordinate
np	Number of points
$nspn$	Number of blade sections
r	Radial coordinate
u	Non-dimensional chordwise coordinate
v	Non-dimensional coordinate perpendicular to u
x	Axial coordinate

Greek

α	Absolute flow angle
β	Relative flow angle
β^*	Blade angle
δ	Deviation angle
ζ	Stagger angle
θ	Tangential direction

Subscripts

b	Blade
$bbot$	Blade bottom
$bstgr$	Blade stagger
$btop$	Blade top
r	Radial
ref	Reference

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x	Axial
s	Streamline
LE	Leading Edge
ND	Non Dimensional
$Norm$	Normal
TE	Trailing Edge
$3D$	Three dimensional

Abbreviations

2D	Two Dimensional
3D	Three Dimensional
3DBGB	Three Dimensional Blade Geometry Builder
CAD	Computer Aided Design
CAPRI	Computational Analysis PRogramming Interface
CFD	Computational Fluid Dynamics
EEE	Energy Efficient Engine
FEA	Finite Element Analysis
IGES	Initial Graphics Exchange Specification
NACA	National Advisory Committee for Aeronautics
NURBS	Non-Uniform Rational B-Splines
UG	UniGraphics-NX

INTRODUCTION

The main property of a blade is to efficiently and smoothly turn the flow and change the fluid velocity. Thus, blade design is a very crucial part of any engine, pump or turbine design process which is iterative in nature. Parametric design of blade shapes is the best approach for such processes as the design cycle can be repeated quickly and easily after every necessary modification to the blade geometry.

There have been several blade design models using parametric approach for blade geometry construction. Koini et al. [1] describes a tool for parametric design of the turbomachinery blades. Using the design parameters, 2D blade sections are created and using NURBS surfaces, 3D blades are obtained. These geometries can be exported to any CAD or mesh generation and analysis software. The stacking of the sections is controlled either by a guide line which is a straight line (common radius) or a 3D NURBS curve that passes through the leading edges or through the centers of gravity of the blade sections as explained in the paper. The meridional profiles are generated with the help of the control points. The tool also has the capability of creating the hub and the shroud surfaces as a single NURBS surface with the help of the control points. The single curve construction approach minimizes the curvature discontinuity errors and the 3D blade generation is simplified as explained in the paper.

Anders et al. [2] described the blade profile using two Bezier curves of order five. The leading and trailing edges were defined using circles or ellipses. Slope continuity was maintained with some curvature discontinuities at certain junctions. The design system consisted of automatic blading which utilized the parametric approach to create certain rules and blades for existing and new compressors like BR715 Booster, HP9 research compressor, Trent500 and Trent800 HP compressor. The next step was optimizing the 2D blade profiles based on the desired aerodynamic properties. A parametric 3D blade manipulation was carried out by 3D blade stacking and radial blade smoothing and interpolation resulting in a smooth 3D blade. This whole process was automated and optimized to generate blade shapes satisfying the required objectives. Usually, Bezier curves of order 3 or higher are used for the mean camber definition to avoid curvature discontinuites and NURBS are used for joining curves as it is the most widely used set of spline curves in many CAD systems. Using coordinates of the control points as parameters proves more useful in obtaining the desired airfoil shape.

Korakianitis [3] describes a method for generating 2D blade shapes. A mixture of analytic polynomials and a desired curvature distribution on a 2D plane is mapped with the blade geometry such that the blade surfaces have continuous slope of curvature throughout their length. The leading edge is defined by 2 thickness distributions around 2 independent construction lines which avoids the overspeed regions near the leading edge. The design method presented in this paper ensures that the boundary layer is not perturbed unintentionally. Effects of stagger angle on blade loading and wake thickness was also explored.

BladeCAD, an interactive geometric design tool was developed by Oliver et al. [4] in 1996 under NASA sponsorship. Using the interface, blade sections were created with respect to general surfaces of revolution and the entire design was defined using a non-uniform rational B-spline (NURBS) surface. An IGES file was the output which is portable to most analysis and design applications. The blade design was defined by constructing the mean camber line, which is a bezier curve of third degree defined by blade angles and the stagger angle. The shape of the mean camber was specified in the angle preserving space $m' - \theta$. The blade section was constructed by defining the suction side and the pressure side curves along with the thickness definition which is normalized by the chord. Using an inverse mapping procedure as explained in the paper, the blade section constructed in $m - r\theta$ space is converted to its respective stream surface in $r - z - \theta$ space. A stacking axis was defined to loft the blade sections in 3D space. The software has an input panel which contains a thickness function editor, section editor which uses NURBS curve to define the mean camber geometry and a blade metal angle editor. It also has the capability of creating general stream surfaces.

This paper presents a parametric design tool for generating 3D blades for various kinds of turbomachinery. A general design methodology is explained in detail which is applicable to a variety of blade designs. The parametric definition of splines through control points makes it very easy to modify the geometry and obtain newer blade shapes without any CAD interference. A desired number of 3D blade sections are obtained which can be lofted into a smooth blade using any CAD package. This has been demonstrated through example cases of EEE High Pressure Compressor blades, wind turbine blades and centrifugal compressor blades. These different types of example cases prove the generality of the design tool. Apart from these designs, the tool is also capable of exploring novel geometry concepts by adding features to the blade shape through spline modification of specific quantities and exploiting the solid-solid union function in CAD. This has also been demonstrated in this paper.

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3D BLADE DESIGN PROCESS

A general method of designing 2D blade sections using the input parameters and stacking them radially to obtain a 3D blade model is explained here. Figure 1 explains the process flow involved. Smooth blades are achieved through robust application of essential geometrical and aerodynamic parameters as input alongwith the streamline coordinates x_s, r_s defined by an axisymmetric run like T-Axi [5–8], or smooth lines defined between the hub and casing. The input also has airfoil stacking information and a control table which defines the sweep, lean and any required quantity with a few control points for adding more definition to the blade geometry. Using UniGraphics (CAD Package), the 3D blade sections are lofted into a solid 3D blade shape. A simple meshing operation creates surfaces through the 3D blade sections to form a solid body. CAPRI [9, 10] is used to parametrically update the spline information on the blade shape to obtain newer blade shapes. The 3DBGB [11] code was built to create the 3D blade shapes presented in this paper and the executable with several cases is available at http://gtsl.ase.uc.edu/3DBGB/.

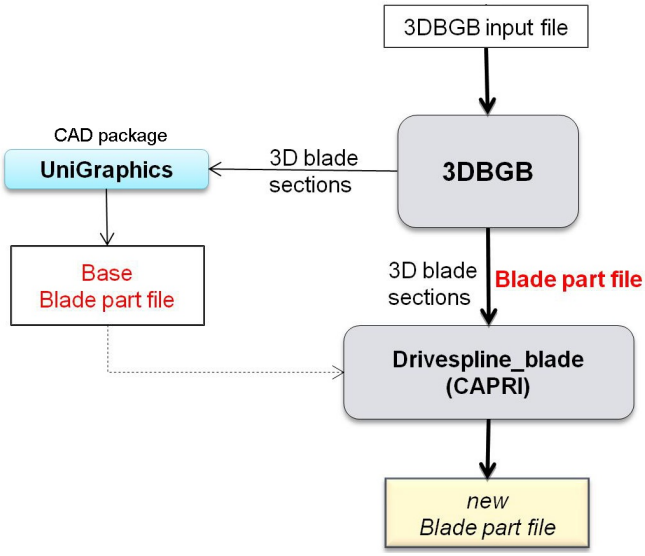


Figure 1. 3D Blade Design process Flowchart.

Governing Equations

The coordinate system used is shown in Figure 2. The meridional view of the streamlines with the leading edge and the trailing edge is shown in Figure 3. The mathematical relations involved in defining the streamline are as below:

1. streamline coordinates: x_s, r_s.
2. airfoil coordinates: m'_b, θ_b.

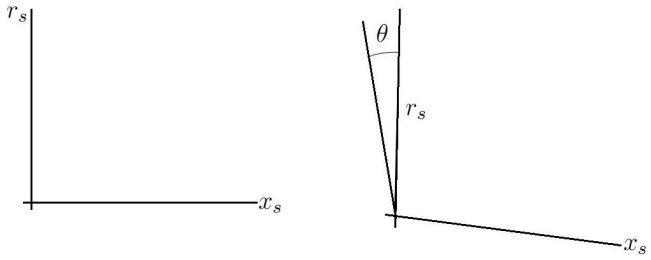


Figure 2. $r-x-\theta$ space for the 3D blade.

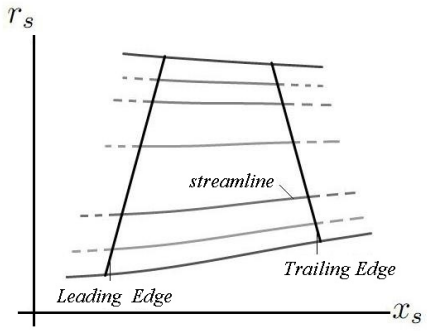


Figure 3. Meridional view of the blade.

The projection of the streamline onto the meridional plane $x-r$ is given by:

$$dm_s = \sqrt{(dr_s)^2 + (dx_s)^2} \tag{1}$$

The normalized differential arc length is defined by:

$$dm'_s = \frac{dm_s}{r_s} \tag{2}$$

The m'_s coordinate of the streamline is obtained by integrating:

$$m'_s = \int \frac{dm_s}{r_s} = \int \frac{\sqrt{(dr_s)^2 + (dx_s)^2}}{r_s} \tag{3}$$

If the airfoil is designed on constant radius sections, then

$$m'_s = \int \frac{dx_s}{r_s} = \frac{1}{r_s}\int dx_s = \frac{x_s}{r_s} \tag{4}$$

which represents a normalized axial coordinate. Using this approach, 3D blades for axial, radial turbomachines and wind turbines have been constructed.

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Blade section construction

A specified number of blade sections stacked radially results in a 3D blade as seen in Figure 4. A blade section contains a mean camber curve passing through the leading edge and trailing edge of the blade section and has a suction side curve and a pressure side curve built around it as shown in Figure 5. Also in Figure 5, the blade metal angle at inlet (leading edge) is defined by β^*_{in} and at the exit (trailing edge) is defined by β^*_{out}. The flow angles at inlet and exit are β_{in} and β_{out} respectively. As shown in this figure, β^*_{in}, β^*_{out} and ζ would all be negative based on the sign convention used. The incidence angle i and the deviation angle δ are given as below:

1. *Incidence Angle*, i: If β^*_{in} greater than zero then $i = \beta_{in} - \beta^*_{in}$ and if β^*_{in} is less than zero then $i = \beta^*_{in} - \beta_{in}$.

2. *Deviation Angle*, δ: If camber is negative then $\delta = \beta_{out} - \beta^*_{out}$ and if camber is positive then $\delta = \beta^*_{out} - \beta_{out}$.

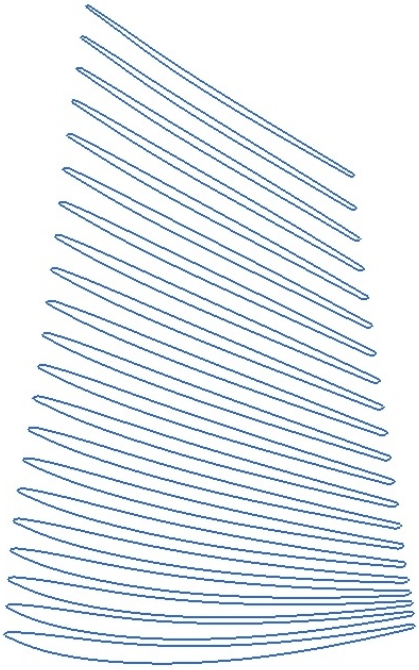


Figure 4. A blade showing the blade sections.

An elliptical leading edge and circular trailing edge are used. The elliptical leading edge helps in keeping the flow attached and laminar [12]. Also, the code has an option of choosing the shape of the leading and trailing edges. The mean camber line is built using the blade angles ($\beta^*_{in}, \beta^*_{out}$), thickness to chord ratio and the meridional chord value. A mixed camber line is defined which is partly cubic in nature. The analytical form of the camber line *cam* is a cubic polynomial as a function of a non-dimensional coordinate u. v is the corresponding non-dimensional coordinate perpendicular to u. The relationship between (m'_b, θ_b) system and (u, v) system is shown in the Figure 6, and where

1. *thk* is the maximum thickness to chord ratio.

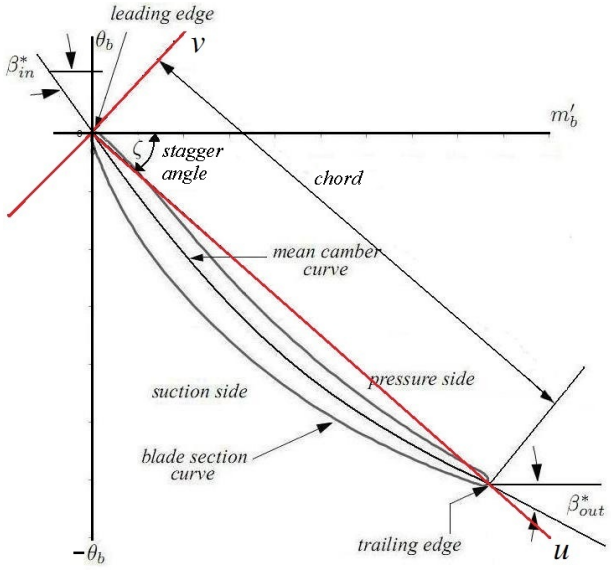


Figure 5. Blade section on the mean camber curve.

2. *chrdx* is the non dimensional meridional chord which is the difference between m'_{TE} and m'_{LE}.
3. In Eq.(11) aa, bb, cc and dd are the coefficients of the cubic equation and ub is the varying parameter which defines the camber line.
4. s_1, s_2 are the slopes at inlet and exit of the blade section.
5. fl_1 = constant slope at inlet as seen in Figure 8.
6. fl_2 = constant slope at exit as seen in Figure 8.
7. c_1, c_2 are the derivatives of u_1, u_2 with respect to angle where u_1, u_2 are defined as given below.

$$s_1 = \tan\beta^*_{in} \tag{5}$$
$$s_2 = \tan\beta^*_{out} \tag{6}$$
$$u_1 = fl_1 \cos\beta^*_{in} \tag{7}$$
$$c_1 = fl_1 \sin\beta^*_{in} \tag{8}$$
$$u_2 = 1 - fl_2 \cos\beta^*_{out} \tag{9}$$
$$c_2 = -fl_2 \sin\beta^*_{out} \tag{10}$$

The analytical form of the camber line *cam* is:

$$cam = aa(ub)^3 + bb(ub)^2 + cc(ub) + dd \tag{11}$$
$$cam_u = 3aa(ub)^2 + 2bb(ub) + cc \tag{12}$$
$$ub = u - u_1 \tag{13}$$
$$dd = c_1 \tag{14}$$
$$cc = s_1 \tag{15}$$
$$aa = \frac{s_1 + s_2 - 2(\frac{c_2 - dd}{xb})}{(xb)^2} \tag{16}$$
$$bb = \frac{-s_1(xb) - aa(xb)^3 + c_2 - dd}{(xb)^2} \tag{17}$$
$$xb = u_2 - u_1 \tag{18}$$

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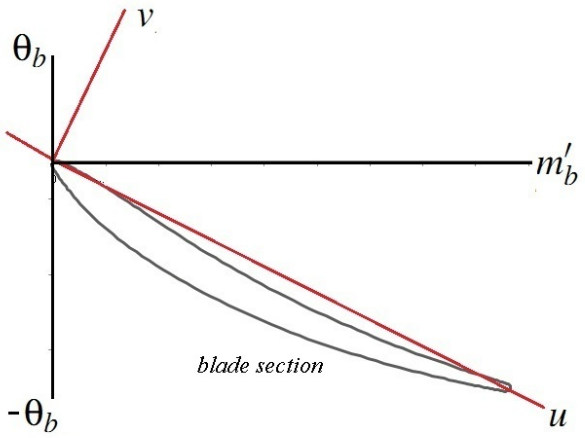


Figure 6. Relationship between (m'_b, θ_b) system and (u, v) system.

The camber line plot of 10th and 11th blade sections for rotor 1 blade of the GE EEE high pressure compressor is shown in Figure 7. Figure 8 shows the plot of camber angle for the same rotor blade sections. Also, the curvature of the camber line for the 10th and 11th blade sections are plotted in Figure 9. All these plots are with respect to u, the non-dimensional coordinate spanning the interval [0,1]. An equally spaced grid of 100 points (can be varied) in this interval was used for the plot.

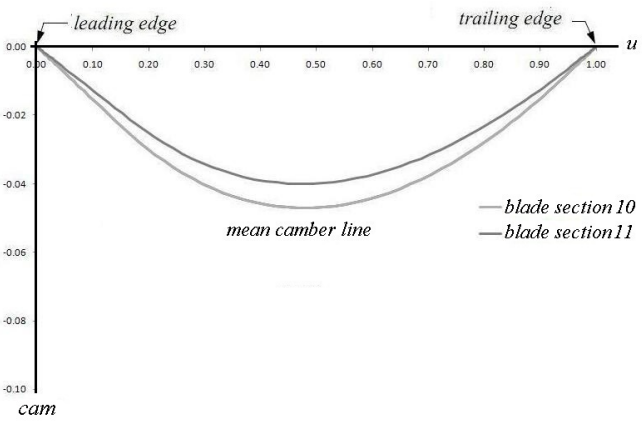


Figure 7. Plot of the camber line of blade sections 10 and 11 of the GE EEE HPC rotor 1 blade.

The camber angle, *ang* is given by differentiating Eq. (11) with respect to *ub* and taking the arctangent of the value obtained.

$$ang = tan^{-1}(cam_u) \tag{19}$$

Using u, cam, thk and ang, the non dimensional coordinates for suction side (u_{btop}, v_{btop}) and pressure side (u_{bbot}, v_{bbot}) are calculated as below:

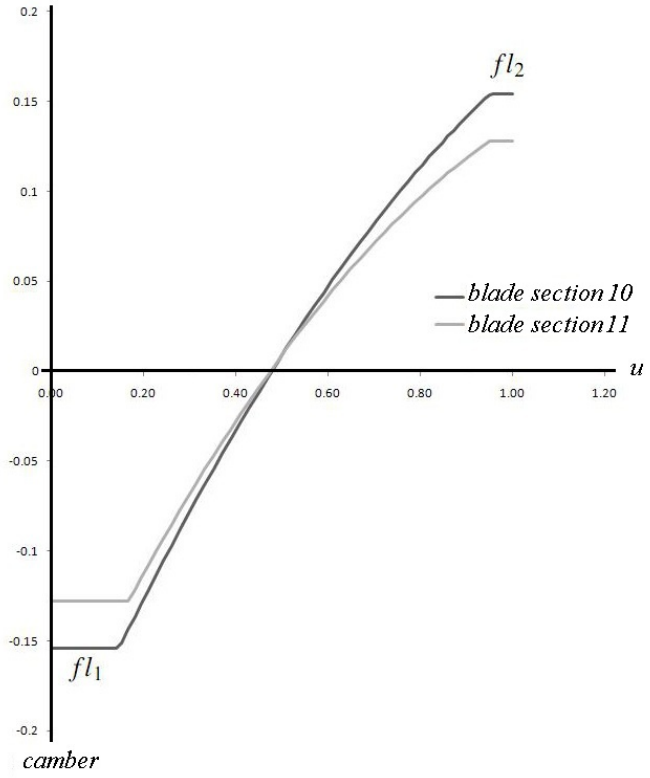


Figure 8. Camber angle plot of blade sections 10 and 11 of the GE EEE HPC rotor 1 blade.

$$u_{bbot} = u + thk\sin(ang) \tag{20}$$
$$v_{bbot} = cam - thk\cos(ang) \tag{21}$$
$$u_{btop} = u - thk\sin(ang) \tag{22}$$
$$v_{btop} = cam + thk\cos(ang) \tag{23}$$

The suction side and pressure side coordinates are merged into a single array of specified number (199 in this case) of non-dimensional coordinates (u_b, v_b) respectively, such that the array starts from the trailing edge and goes counter-clockwise through the leading edge and comes back to the trailing edge. Since the leading edge and trailing edge coordinates of the pressure side and suction side are the same, only a single value of the leading edge coordinate is used in the array and the first and last point in the array is the same trailing edge coordinate. The non dimensional leading edge is chosen as (0,0) and trailing edge as (1,0) for convenience. The stagger angle, ζ is calculated by taking the average of the flow angles at inlet and exit. The normalized staggered airfoil coordinates $(m'_{bstgr}, \theta_{bstgr})$ are as given below:

$$\zeta = (\beta^*_{in} + \beta^*_{out})/2 \tag{24}$$
$$m'_{bstgr} = u_b\cos(-\zeta) + v_b\sin(-\zeta) \tag{25}$$
$$\theta_{bstgr} = -u_b\sin(-\zeta) + v_b\cos(-\zeta) \tag{26}$$

Figure 10 shows the airfoil rotated by the stagger angle and is placed on the unit chord on the non-dimensioanl spacing array

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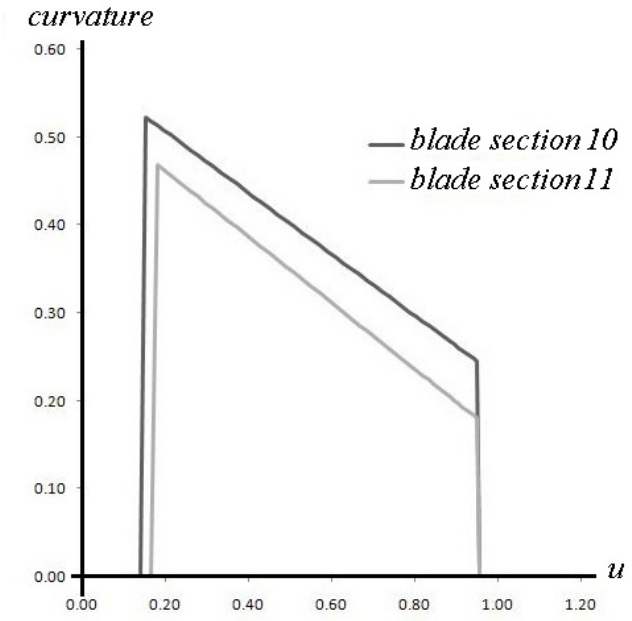


Figure 9. Plot showing the curvature of the camber line of blade sections 10 and 11 of the GE EEE HPC rotor 1 blade.

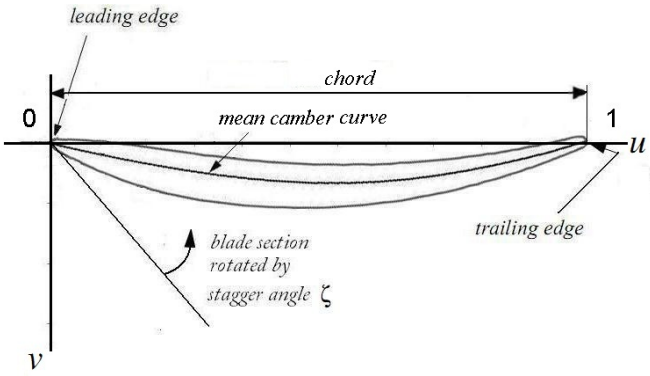


Figure 10. Blade section on the normalized unit chord.

system (u,v). The actual chord is calculated and the airfoil coordinates are scaled by multiplying the staggered coordinates $(m'_{bstgr}, \theta_{bstgr})$ with the non-dimensional actual chord as below:

$$chrdx = \left|m'_{TE} - m'_{LE}\right| \tag{27}$$

$$chrd = chrdx/\left|\cos\zeta\right| \tag{28}$$

$$m'_b = m'_{bstgr}(chrd) \tag{29}$$

$$\theta_b = \theta_{bstgr}(chrd) \tag{30}$$

The resultant 2D blade section contains np values of m'_b, θ_b coordinates in the meridional coordinate system.

Variety of Airfoil shapes

3DBGB is flexibile in its capability of generating different airfoil shapes to cope with a variety of blade designs. Since each blade section is constructed independently through parameters, it is possible to have different blade sections at different spanwise locations. For example, a wind turbine blade has a circular hub and progresses towards an S809 airfoil shape radially. The general capability of the blade section construction makes it easy to deal with such shapes. Currently, 3DBGB is capable of generating the S809 airfoil, NACA 4-digit airfoils, a circular section, and the default airfoil which is a mixed camber airfoil as shown in Figure 11. In addition to these, the tool is flexible in adding various airfoil shapes parametrically. It is recognized that the default blade section is inadequate for a transonic fan, but those sections can be added in a parametric fashion. This is also true for many other blade types. The integrated ability for a general blade section and consistent 3D stacking in a rigorous manner sets this blade generator apart.

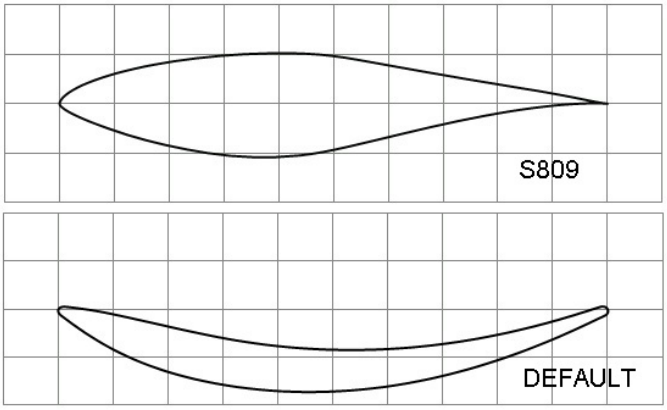


Figure 11. Variety of airfoils generated by 3DBGB in addition to circular and 4 digit NACA airfoils.

Streamwise Mapping of the Blade Sections

The streamlines are defined using x_s and r_s coordinates, both upstream and the downstream of the blade for robust mapping. Specific number of streamlines pass through a blade geometry. Typical streamsurfaces defined by revolving the streamlines about the x-axis for an axial machine is shown in Figure 12 and for a radial machine is shown in Figure 13. The 2D blade sections created are mapped with their corresponding streamsurfaces (surfaces of revolution of 2D streamline curves).

A cubic spline is used for the streamline coordinates x_s, r_s with m'_s as the spline parameter. m'_{sLE}, the leading edge m'_s value is calculated by taking the inverse spline of $x(m'_s)$ evaluated at x_{LE} on each of the streamline as shown in Figure 14. Similarly, inverse spline of $r(m'_s)$ will be evaluated at r_{LE} on each of the streamline for obtaining m'_{sLE} value of a radial leading edge. The meridional offset between the blade leading edge m'_{bLE} and the streamline leading edge m'_{sLE} is calculated and is necessary for precise conformation of the blade section on the corresponding streamline. This is because the zero m'_s on each streamline is different from the m'_{bLE} on each corresponding blade section and this offset is expressed as:

$$\delta m' = m'_{sLE} - m'_{bLE} \tag{31}$$

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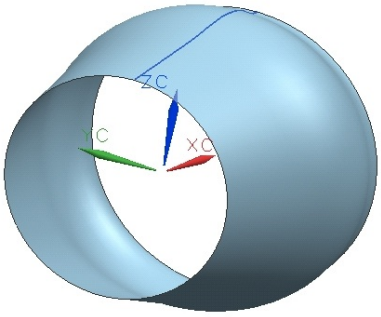


Figure 12. Stream surface for an axial machine.

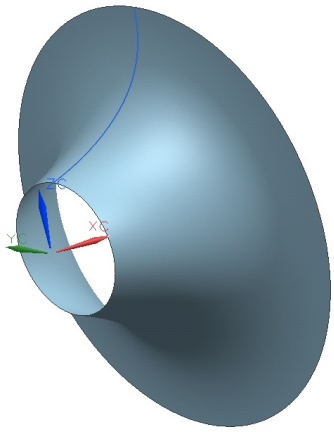


Figure 13. Stream surface for a radial machine.

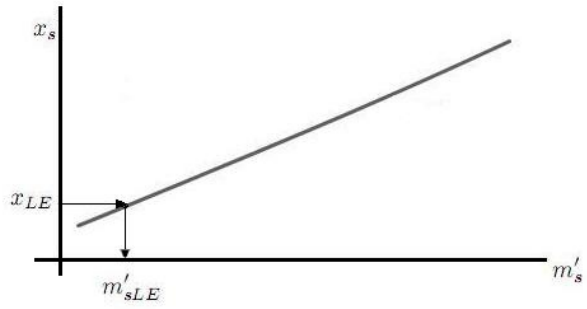


Figure 14. m'_{sLE} obtained using inverse spline on m'_s.

Similarly, there exists a tangential offset which is added to the θ_b coordinates to obtain streamwise θ_{3D} coordinates for the blade:

$$\theta_{3D} = \theta_b + \delta\theta \tag{32}$$

Once all the offsets are calculated, the streamwise meridional coordinates m'_{3D} for each blade section is obtained using:

$$m'_{3D} = m'_b + \delta m' \tag{33}$$

The m'_{3D} coordinates are used to calculate the streamwise x_{3D} and r_{3D} coordinates by evaluating the spline at every streamwise meridional coordinate as below:

1. x_{3D} = spline evaluated at each m'_{3D} for all the np points of each blade section.
2. r_{3D} = spline evaluated at each m'_{3D} for all the np points of each blade section.

The spline evaluation process is depicted in Figures 15 and 16.

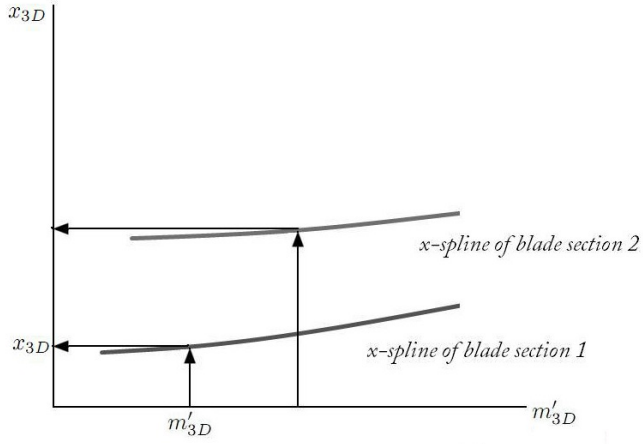


Figure 15. x_{3D} obtained by spline evaluation at each m'_{3D} of each blade section.

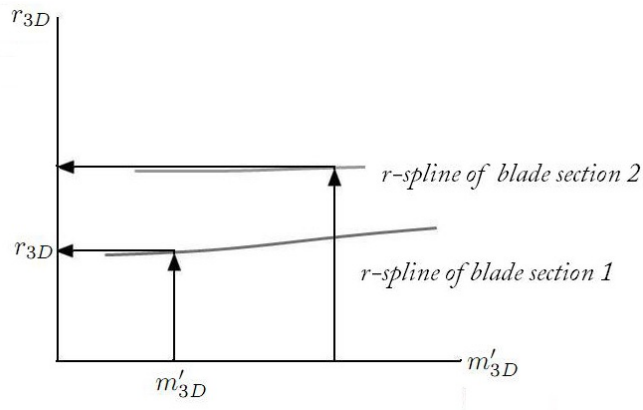


Figure 16. r_{3D} obtained by spline evaluation at each m'_{3D} of each blade section.

x_{3D}, r_{3D} and θ_{3D} coordinates for all the $nspn$ 3D blade sections are thus calculated in the cylindrical coordinate system.

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Sweep and Lean to the Blade

Spanwise Sweep and Lean are defined by a specific number of control points (span,$\delta m'$) and (span,$\delta\theta$). The control points are used to construct a smooth sweep and lean definition spanwise by B-splines. The normalized span of nth streamline is given by :

$$L_{ref} = r_{LETIP} - r_{LEHUB} \tag{34}$$

$$span(n) = r(n) - r_{HUB} \tag{35}$$

$$\widetilde{span(n)} = span(n)/(L_{ref}) \tag{36}$$

where L_{ref} is the blade height.

The corresponding $\delta m'$ values at each streamline are evaluated by iterpolating the $\delta m'$ curve at each spanwise location on each streamline as depicted in Figure 17. In this manner, the sweep is added to the blade whenever required. Figure 18 shows blades with forward and backward sweep respectively.

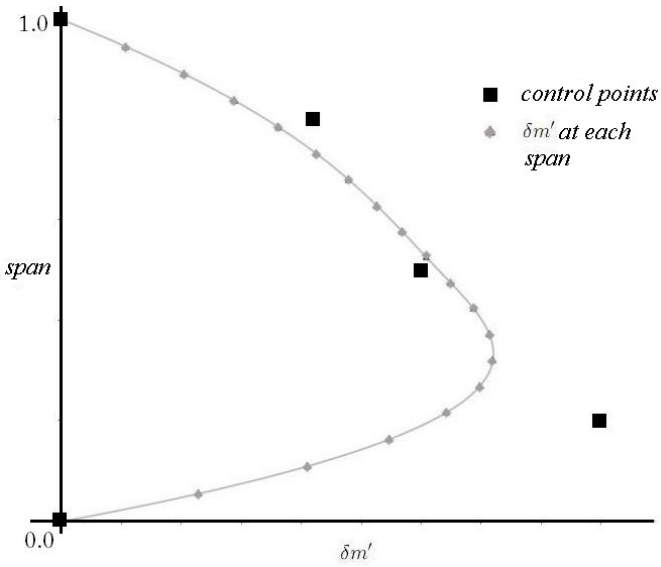


Figure 17. $\delta m'$ evaluated at each span.

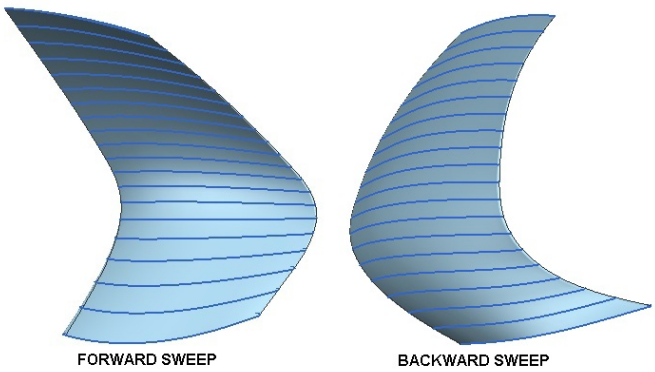


Figure 18. Forward and backward swept blades respectively.

Similarly the lean is added to the blade as required by defining the lean with few control points and obtaining the $\delta\theta$ values at each streamline as explained above. This method as shown in Figure 19 was used to modify the lean near the hub of rotor 3 of booster optimum shown as point 2 in the paper by Park et. al [13] to eliminate the corner separation at the exit which increased the adiabatic efficiency for the isolated blade row 3D CFD analysis of rotor 3 as explained by Siddappaji et. al [11]. The lean modification at the hub of the rotor 3 for the redesigned booster is shown in Figure 20.

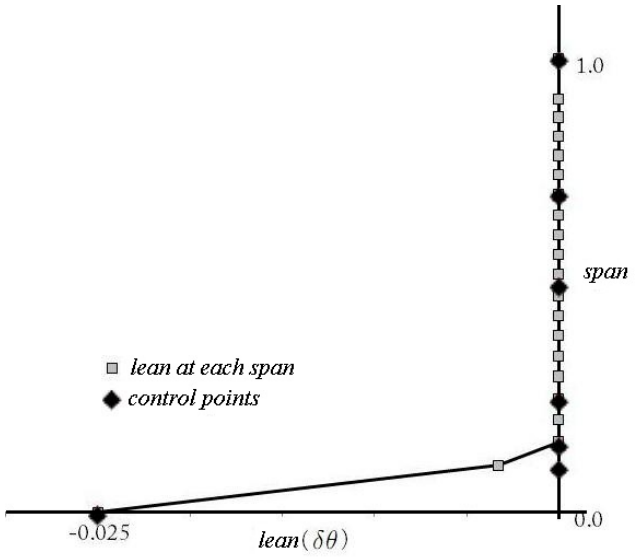


Figure 19. Spanwise lean modification on rotor 3 of a booster optimum.

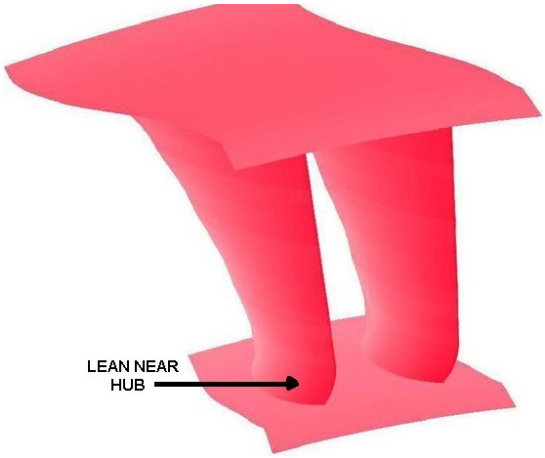


Figure 20. Lean added near the hub on rotor 3 of a booster.

Radial Stacking of 3D Blade Sections

The 3D blade sections are stacked radially by including a θ offset, θ_{stack} for each individual blade section. Stacking is defined as a fraction of the non-dimensional actual chord. This

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value is transformed into the average θ_{3D} coordinate which is subtracted from the array of θ_{3D} coordinates to obtain the stacking. The stacking can be done at the leading edge, trailing edge and at any percentage of the chord.

$$\theta_{3D} = \theta_b + \delta\theta + \theta_{stack} \tag{37}$$

Figure 21 explains the theta offset defined for stacking the blade at the leading edge, percentage chord and the trailing edge correspondingly. Finally, using the coordinate transformation, the

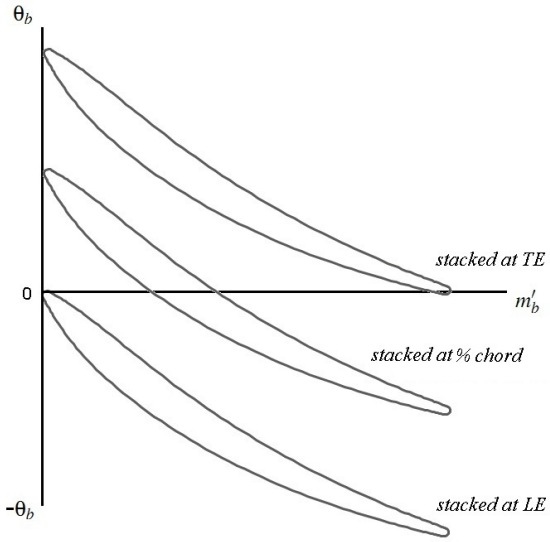


Figure 21. θ offset defined for various stacking option.

3D blade coordinates are converted from the cylindrical system to the cartesian system in which the engine axis is assumed to be along the X-axis as given below:

$$x_{3D} = x_{3D} \tag{38}$$
$$y_{3D} = r_{3D} \times \sin\theta_{3D} \tag{39}$$
$$z_{3D} = r_{3D} \times \cos\theta_{3D} \tag{40}$$

Output and 3D Blade CAD Model

3DBGB outputs *nspn* data files, each containing *np* values of 3D coordinates for all the *nspn* blade sections. These 3D blade section files can be imported into any CAD package to generate a 3D blade CAD model. A lofting procedure is performed to create 3D blade surfaces passing through the blade sections. Figure 22 shows a smooth 3D blade obtained by lofting in Unigraphics (CAD package) where all the imported blade sections are used.

Newer Blade Shapes using CAPRI

CAPRI [9, 10] (*Computational Analysis PRogramming Interface*) is used to obtain newer blade shapes using the 3D blade constructed in UG (CAD package) as a base model. This is achieved by parametrically updating the spline information of

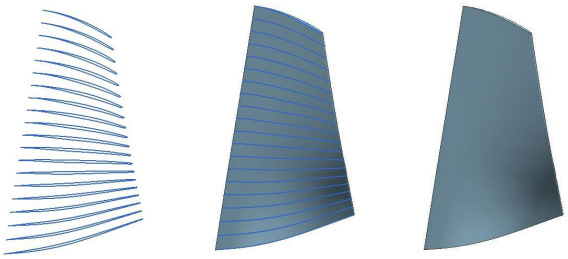


Figure 22. 3D Blade lofted in UG.

each 3D blade section on the base model through a script in C integrating the CAPRI interface and the base model. The new spline information is the parameter to be updated on the base model. This approach speeds up the process of modifying the geometry without going into the CAD interface everytime a change is required and avoids geometry translation errors as the morphing of the blade geometry is done parametrically.

BLADE SPECIAL FEATURE CAPABILITY

Additional features can be added to the blade such as extrusion and fillets. Concepts like bent tip and split tip can be constructed by modifying the blade shape through control points. A smooth blade with these additional features can be generated using the lofted blades and the solid-solid union function in CAD to create a solid blade. Below are some of the features explained and this capability can be extended to explore feasible blade geometry concepts. The following examples depict the extreme capability of exploring such novel concepts.

Extruded Blade

In some cases, grid generators require an extruded blade due to tolerance issues and extrusion is a property which is useful for other purposes as well. It is achieved through a simple offset of the hub and the tip streamline coordinates in the normal direction to the streamline. At any point m'_s, the normal in the x-direction, x_{NORM} and the normal in the r-direction, r_{NORM} are calculated by using the orthogonal property between the normal and the slope at that point.

$$x_{NORM} = \frac{dr_s}{dm'_s} \tag{41}$$
$$r_{NORM} = -\frac{dx_s}{dm'_s} \tag{42}$$

So, the offset in the normal direction as shown in Figure 23 is given by

$$\Delta n = aL_{ref}, \tag{43}$$

where a is the percent offset desired and L_{ref} is the reference length. Also,

$$\Delta n = b\sqrt{x_{NORM}{}^2 + r_{NORM}{}^2} \tag{44}$$

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where b is an intermediate extruded blade scale factor. b is solved

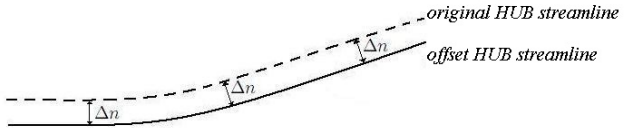


Figure 23. Offset at the hub streamline.

using the 4 equations above, and the offset at the hub is obtained:

1. streamline coordinates at the hub : x_{sHUB}, r_{sHUB}
2. streamline coordinates at the tip : x_{sTIP}, r_{sTIP}

$$x_{sExtruded} = x_{sHUB} + (b \times x_{NORM}) \quad (45)$$
$$r_{sExtruded} = r_{sHUB} + (b \times r_{NORM}) \quad (46)$$

and the offset at the tip as

$$x_{sExtruded} = x_{sTIP} - (b \times x_{NORM}) \quad (47)$$
$$r_{sExtruded} = r_{sTIP} - (b \times r_{NORM}) \quad (48)$$

The new extruded hub and tip streamline coordinates are used instead of the original coordinates and the blade parameters corresponding to the original hub and tip streamlines are used as the input to obtain an extruded 3D blade as shown in Figure 24 using the procedure as explained before.

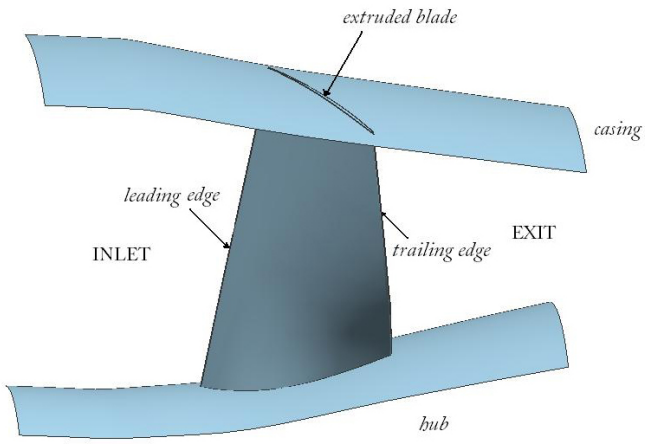


Figure 24. Extruded blade at the tip with hub and casing.

Fillets

Generally, 3D blades contain fillets at the hub to decrease the blade stresses and have a steady flow which reduces the flow separation near the hub. Using UG (CAD package), fillets can be easily added to the 3D blade model generated as shown in Figure 25. An edge blend operation creates the fillet at the intersection of the hub blade section and the hub surface. With CAPRI, the radius of the fillets are treated as a parameter and can be updated to obtain newer fillet sizes on blade shapes without going back into UG to modify the fillet. Figure 26 shows the fillet radius updated from 2 mm to 5 mm on the same blade using CAPRI. This capability saves time and is useful in an automated chain.

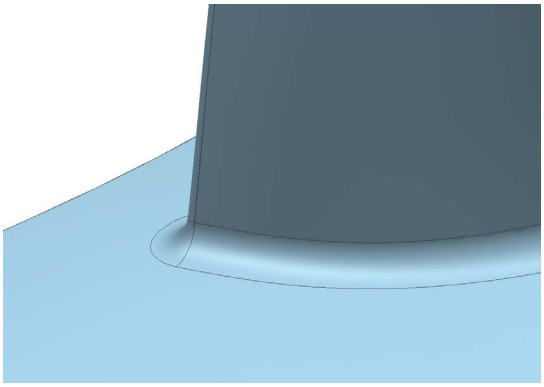


Figure 25. Fillet definition on a blade near the hub.

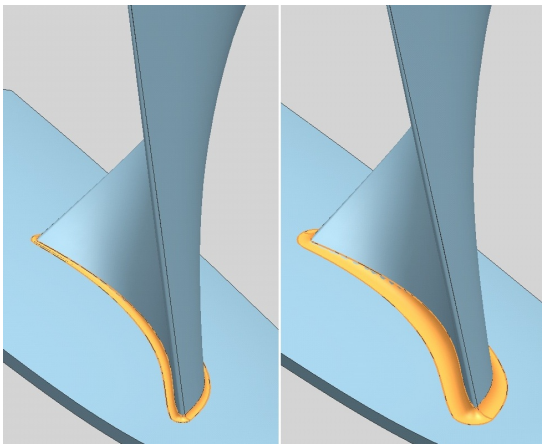


Figure 26. Fillet radius updated from 2 mm to 5 mm on a blade using CAPRI.

Blades with Bent Tip

Some novel concepts such as blade shapes with some amount of lean near the tip can be explored using this tool. Since the lean is defined by few control points spanwise, the lean can be modified near the tip keeping zero lean on the rest of the blade span as shown in Figure 27. This results in a blade shape with

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bent tip as shown in Figure 28 which is rendered using UG after importing the blade sections with the tip section leaned. Such concepts can be explored with quick and easy modification of the blade shape due to the parametric nature of the spline defining the geometry.

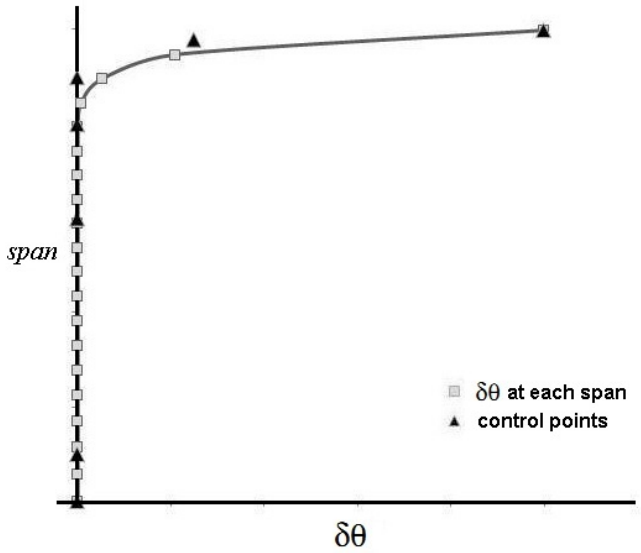


Figure 27. Plot showing the tangential lean near the tip with control points.

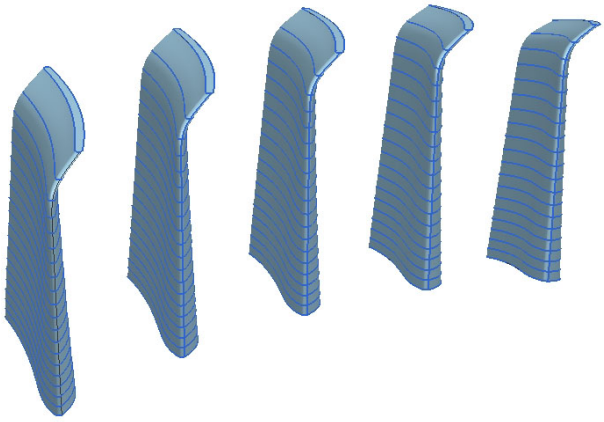


Figure 28. Bladerow with bent tip.

Blades with Split Tip

Another novel concept which is explored is a blade with a split tip. The blade tip is split and one part is leaned positively and the other part negatively. Due to the parametric definition of the spline through few control points it is easy to modify the lean in both directions. A blade section near the tip has a constriction near the mid chord which acts as the base blade section for the split tip blade sections to be imported and lofted in UG as shown in Figure 29. The meshing and joining the split tip part with the rest of the blade is done in UG. In this manner a smooth blade shape with a split tip can be obtained.

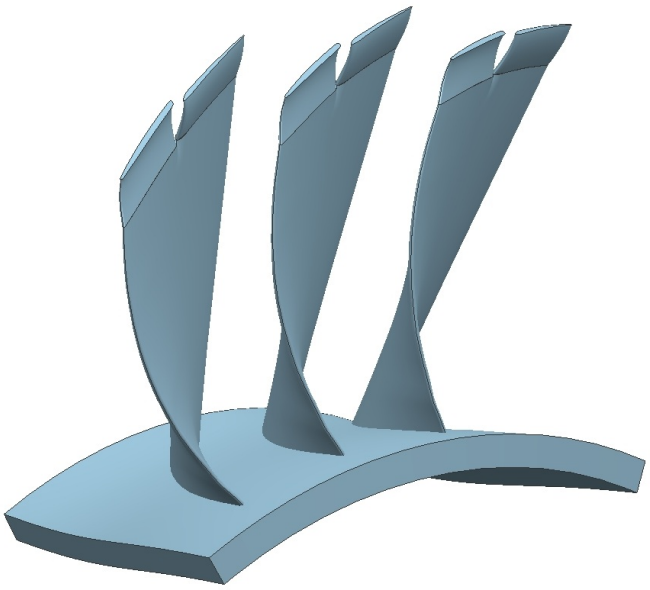


Figure 29. Bladerow with split tip.

Blades with multiple Split Tips

The alula [14] is a small projection on the anterior edge of the wing of modern birds as shown in Figure 30. During flying at low speeds or landing, when the alula is moved upward and forward it allows the wing to achieve a higher than normal angle of attack which results in lift without any stall [15]. To simulate this effect on aircraft wings, tip sails are built at the tip of the wing as shown in Figure 31 and the benefits are explained in the book by Hoerner [16]. It is expected that this design can take advantage of a fresh leading edge boundary layer and second tip vortex. It is not yet known if this design has any advantage. It should be recognized that it may create higher stresses and the design would need to be part of a multidisciplinary optimization including stress analysis.



Figure 30. Alula on the wing of an Adult Bald Eagle. [14]

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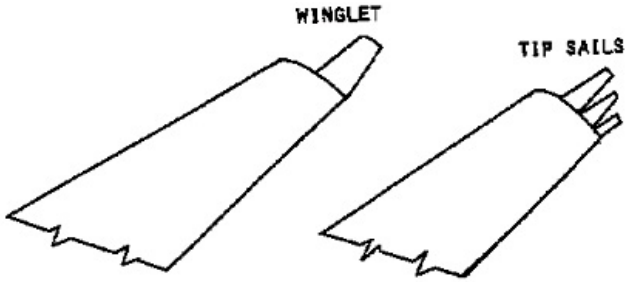


Figure 31. Wing tip vortex drag recovery devices. [16]

Blades with 2 tips leaned in opposite Directions

Another novel concept explored is a blade having 2 tips with one tip leaned in one direction and the other tip in the other direction. The blade was generated once with positive lean near the tip and also with negative lean at the tip using the spline modification explained earlier. Using UG the 2 blades were merged to obtain the desired blade shape as shown in Figure 32. The blade sections with zero lean are alligned and the resultant blade has 2 tips leaned in opposite directions.

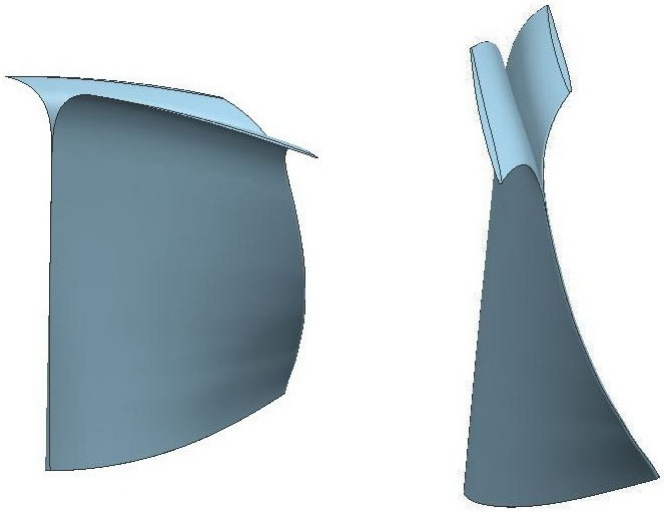


Figure 32. Blade with 2 tips leaned in opposite directions.

Endwall Tandem Blades

Figure 34 shows the tandem blade concept and the detailed numerical investigation of tandem airfoils for a core compressor rotor is explained in the paper by McGlumphy et. al [17]. Endwall Tandem Blades are obtained by merging two similar blades, each swept in opposite directions. Two blades are created with forward sweep and backward sweep respectively and using UG they are merged to obtain the desired blade shape as shown in Figure 33. The blade sections with zero sweep are alligned and the resultant blade shape has tandem rows at the hub and the tip.

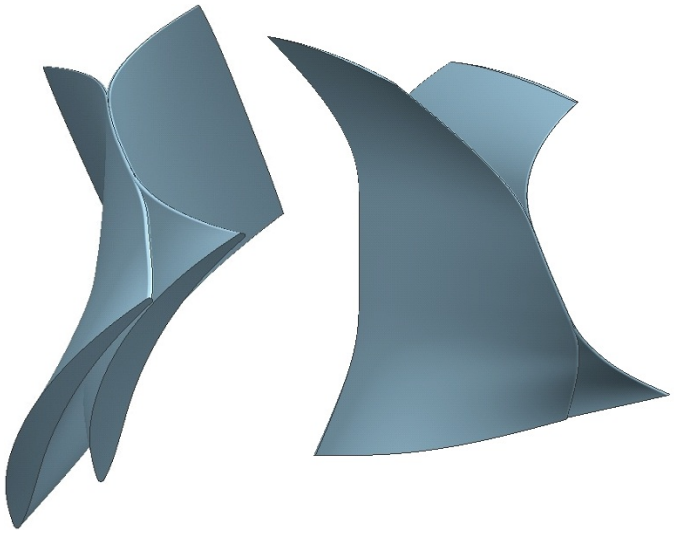


Figure 33. Blade with tandem rows.



Figure 34. Tandem Airfoil profile and a single airfoil profile comparison. [17]

EXAMPLE DESIGN PROBLEMS

The robustness of the design tool is demonstrated through the following examples. An axial compressor blade, a wind turbine blade and a radial compressor blade were created.

EEE High Pressure Compressor Blade

The EEE High Pressure Compressor has 10 stages and an Inlet Guide Vane (IGV) thus having 21 bladerows. The blade shapes have been created with the data obtained from the NASA report [18]. The report contains airfoil-geometry data such as the metal angles at inlet and exit, maximum thickness to chord ratios, axial and radial coordinates at the leading edge and trailing edge for 12 streamlines. These values were extrapolated to 21 streamlines and used as the input for 3DBGB to obtain the 3D blade shapes. Figure 35 shows the rotor 1 blade row and Figure 36 shows the complete assembly.

Wind Turbine Blade

A wind turbine blade was generated as shown in Figure 37 to demonstrate the flexibility of the design tool. Figure 38 shows

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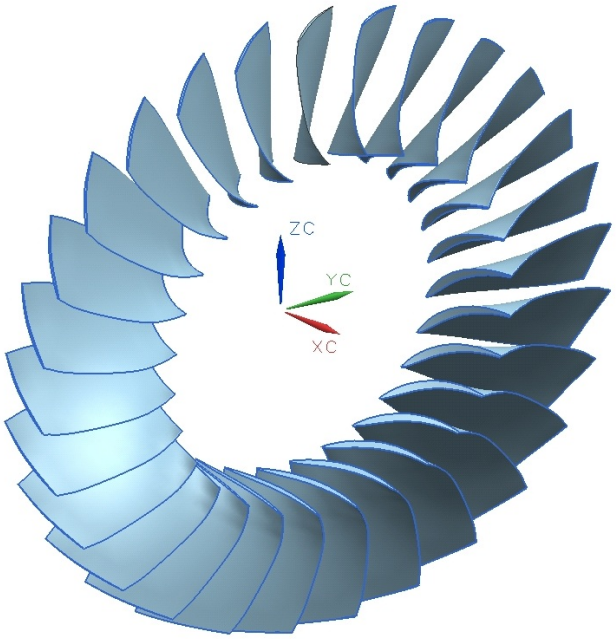


Figure 35. EEE HPC Rotor 1 bladerow.

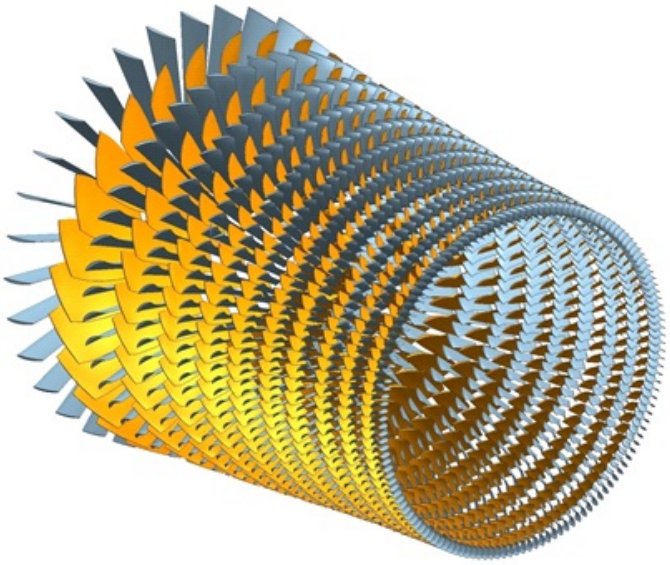


Figure 36. EEE HPC complete assembly.

the wind turbine assembly. One of the issues encountered was going beyond axial ζ equal to 92 degrees where the blade shapes failed to be generated since the *chrd* value calculated according to the Eq. (28) became a non-defined value as cosine of the stagger angle reaches zero. To overcome this issue, a switch was added in the input file which would read the actual non-dimensional chord values *chrd* for such high staggered cases instead of using Eq. (28). The addition of this option expands the capability of the tool to include blades with high stagger.

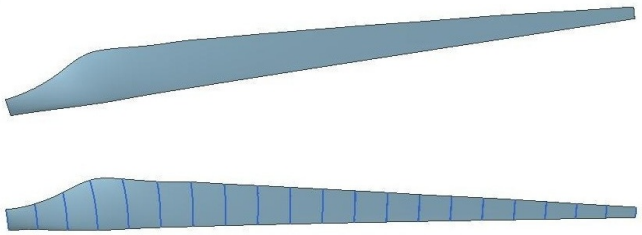


Figure 37. A wind turbine blade.

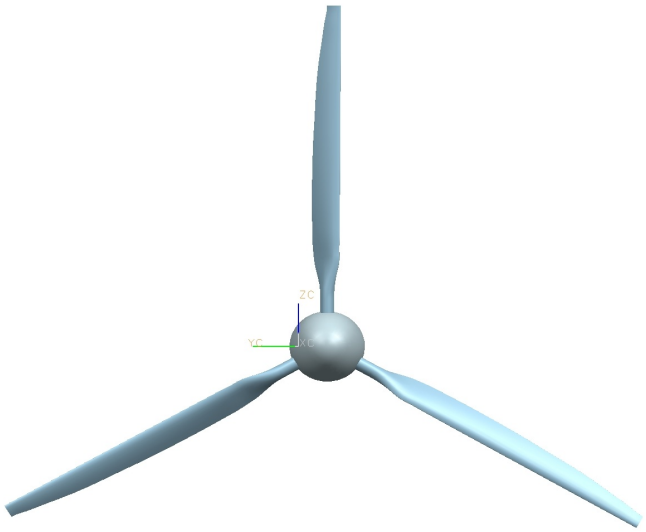


Figure 38. Wind Turbine assembly.

Centrifugal Compressor Blade

To demonstrate the generality of the design tool, the blade shape similar to the Low Speed Centrifugal Compressor [19] was generated. The NASA report [19] contains hub and tip streamline coordinates and other airfoil geometry data. The streamlines were interpolated to obtain 5 streamlines including the hub and tip streamlines. The inlet angle was taken as zero and the outlet angle as 90 degrees. The same design methodology used to create axial blades were used except the blade was stacked at the trailing edge. Also a condition was added to use normalized r_s values to obtain m'_{TE} through inverse spline to calculate the non-dimensional meridional chord when

$$\frac{dr_s}{dm'_s} > \frac{dx_s}{dm'_s} \tag{49}$$

Figure 39 shows the centrifugal compressor assembly.

CONCLUSIONS & FUTURE WORK

A parametric design tool for 3D blade design of turbomachinery has been presented in this paper. The capability to construct 3D blades for all types of turbomachinery has demonstrated the generality of the approach through various example cases. The tool outputs the 3D blade in a specified number of data files containing *np* coordinates of 3D blade sections which

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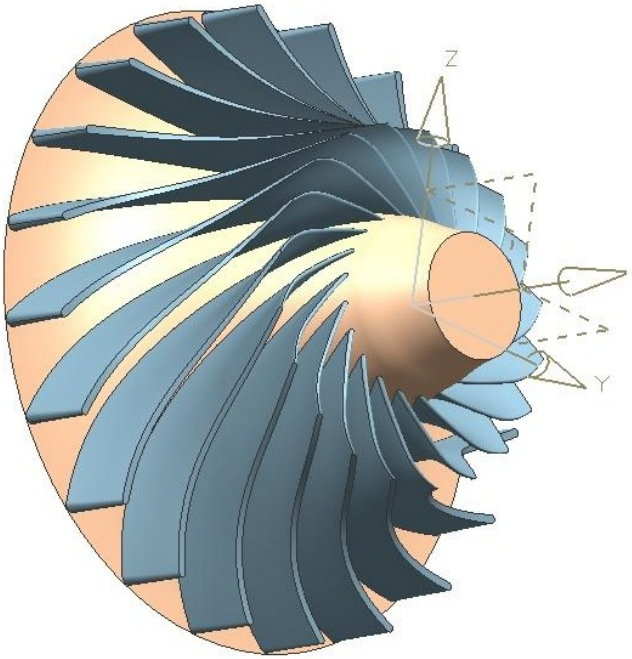


Figure 39. Centrifugal Compressor Blade Assembly.

can be imported into a CAD package to obtain a smooth lofted blade. The parametric definition of the splines makes the geometry modification process quicker and easier and also allows the exploration of some novel concepts. The executable that creates the blades is freely available at http://gtsl.ase.uc.edu/3DBGB/.

Future work includes more section options and defining the 2D blade sections parametrically as a combination of many different sections. The focus will also be on connecting the tool to an optimizer followed by an automated CFD and FEA analysis.

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